

# Binary polynomials vanishing to high order off the origin: the minimum degree at every origin order

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With substantial assistance from Claude and GPT models.  
Details at the end of the main text.

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## Abstract

For  $0 \leq \ell < k$ , let  $\delta(n, k, \ell)$  be the least degree of a polynomial  $P \in \mathbb{F}_2[x_1, \dots, x_n]$  that vanishes with (Hasse) multiplicity at least  $k$  at every nonzero point of  $\mathbb{F}_2^n$  and with multiplicity exactly  $\ell$  at the origin. We prove that for all  $n \geq 1$ ,

$$\delta(n, k, \ell) = n + 2\ell + \sum_{j=0}^{n-1} \left\lfloor \frac{k - \ell - 1}{2^j} \right\rfloor.$$

Menezes determined  $\delta$  over every field for  $k \geq 2$  and  $n \geq k - 1$ ; in characteristic two his value is  $n + 2k - 2 - s_2(k - \ell - 1)$ , which our formula recovers whenever  $k - \ell - 1 < 2^n$ . When  $k - \ell - 1 \geq 2^n$  the value is smaller: each block of  $2^n$  in  $k - \ell - 1$  is paid for by the square of the product of the  $2^n - 1$  affine hyperplanes  $u \cdot x = 1$ ,  $u \neq 0$ , which avoid the origin. The lower bound expands a polynomial in the basis  $x^\varepsilon(x^2 + x)^\varepsilon$ , inverts  $x^2 + x$  locally by an additive power series, and turns the multiplicity conditions into additive forms that vanish on the span of  $x_1^2 + x_1, \dots, x_n^2 + x_n$ ; reducing modulo the vanishing polynomial of this span isolates the lowest homogeneous part of the polynomial composed with that series, which the origin order makes nonzero. Minimizing over  $\ell$  gives the least degree  $D(n, k)$  of a polynomial with multiplicity at least  $k$  off the origin and less than  $k$  at the origin:  $D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor$  for  $k < 2^{n-1}$ , and the multiplicity Schwartz–Zippel bound  $2k - \lfloor k/2^{n-1} \rfloor$ , asked for by Bishnoi, Boyadzhyska, Das and Mészáros, for  $k \geq 2^{n-1}$ . That bound is attained if and only if  $k \geq 2^{n-2}$ . We also give a second proof of  $D(n, k)$ , from the Schwartz–Zippel lemma and a one-dimension reduction. Every mathematical claim of the paper is formalized in Lean 4, and an appendix maps each to its formal proof.

## 1 Introduction

Polynomials that vanish to high order on a grid are a basic tool of the polynomial method. Over the Boolean cube a natural extremal question asks how small the degree of such a polynomial can be when it is required to vanish to order  $k$  everywhere except at one point. Over  $\mathbb{R}$  this question was answered by Sauer and Wigderson [SW20]: the least degree of  $P \in \mathbb{R}[x_1, \dots, x_n]$  with  $P(\mathbf{0}) \neq 0$  and zeros of multiplicity at least  $k \geq 2$  at all other points of  $\{0, 1\}^n$  is  $n + 2k - 3$  when  $n \geq 2k - 3$  [SW20, Theorem 1.3]. They observed that the answer depends on the field and that

characteristic two behaves differently. Menezes [Men26] determined the least degree over every field when  $k \geq 2$  and  $n \geq k - 1$ , also prescribing the order of vanishing at the origin.

This paper determines the binary case in every dimension. Throughout, multiplicity means Hasse multiplicity of formal polynomials (Definition 2.1); in particular no polynomial is reduced modulo  $x_i^2 - x_i$ .

**Definition 1.1.** Let  $n \geq 1$  and  $k \geq 1$ . For  $0 \leq \ell < k$ , let  $\delta(n, k, \ell)$  be the least total degree of a polynomial  $P \in \mathbb{F}_2[x_1, \dots, x_n]$  with  $\text{mult}_a(P) \geq k$  for every  $a \in \mathbb{F}_2^n \setminus \{\mathbf{0}\}$  and  $\text{mult}_{\mathbf{0}}(P) = \ell$ ; we call  $\text{mult}_{\mathbf{0}}(P)$  the *origin order* of  $P$ . A polynomial is *k-admissible* if  $\text{mult}_a(P) \geq k$  for every nonzero  $a$  and  $\text{mult}_{\mathbf{0}}(P) \leq k - 1$  [Lean], and  $D(n, k)$  is the least total degree of a *k-admissible* polynomial. Thus  $D(n, k) = \min_{0 \leq \ell < k} \delta(n, k, \ell)$ .

The condition at the origin excludes  $P = 0$  [Lean] and allows any origin order below  $k$ , not only a nonzero value at the origin. Bishnoi, Boyadzhiyska, Das and Mészáros [BBDM23, Question 4.3] asked, in the study of subspace coverings with multiplicities, whether  $D(n, k) \geq 2k - \lfloor k/2^{n-1} \rfloor$  when  $k \geq 2^{n-2}$ . This bound follows for all  $n$  and  $k$  from the multiplicity Schwartz–Zippel lemma of Dvir, Kopparty, Saraf and Sudan [DKSS13]; see Proposition A.2, which was the content of an earlier note [Note]. Bishnoi et al. also construct multisets of  $2k - \lfloor k/2^{n-1} \rfloor$  affine hyperplanes that cover every nonzero point at least  $k$  times and the origin at most  $k - 1$  times, for  $k \geq 2^{n-2}$  [BBDM23, Theorem 1.2(a) of arXiv v1], and the bound is attained for  $k \geq 2^{n-2}$  (Corollary 1.3).

## 1.1 Main results

For integers  $n \geq 1$  and  $0 \leq \ell < k$  put

$$\Phi(n, k, \ell) = n + 2\ell + \sum_{j=0}^{n-1} \left\lfloor \frac{k - \ell - 1}{2^j} \right\rfloor.$$

**Theorem 1.2.** For all integers  $n \geq 1$  and  $0 \leq \ell < k$ ,  $\delta(n, k, \ell) = \Phi(n, k, \ell)$ : some polynomial with multiplicity at least  $k$  at every nonzero point of  $\mathbb{F}_2^n$  and exactly  $\ell$  at the origin has degree  $\Phi(n, k, \ell)$ , and every such polynomial has degree at least  $\Phi(n, k, \ell)$  [Lean].

Write  $k - \ell - 1 = q2^n + r$  with  $0 \leq r < 2^n$ , and let  $s_2$  be the binary digit sum. By Legendre’s formula (Lemma 3.2),

$$\Phi(n, k, \ell) = n + 2k - 2 - 2q - s_2(r)$$

[Lean]. In particular  $\Phi(n, k, \ell) = n + 2k - 2 - s_2(k - \ell - 1)$  when  $k - \ell - 1 < 2^n$  [Lean]. When  $k - \ell - 1 \geq 2^n$ , that is  $q \geq 1$ ,  $\Phi$  is strictly smaller, since  $s_2(k - \ell - 1) = s_2(q) + s_2(r) \leq q + s_2(r) < 2q + s_2(r)$  [Lean]. A polynomial of degree  $\Phi$  is  $y_1^\ell F^{2q} g_{r+1}$ , where  $y_1 = x_1^2 + x_1$ ,  $F$  is the product of the  $2^n - 1$  affine forms  $1 + u \cdot x$  with  $u \neq \mathbf{0}$ , and  $g_{r+1}$  is a truncated product (Theorem 3.7); its degree is exactly  $\Phi$ . The simplest example with  $k - \ell - 1 \geq 2^n$  is  $n = 1$ ,  $k = 3$ ,  $\ell = 0$ : there  $\delta = \Phi = 3$ , attained by  $(1 + x)^3$ , while  $n + 2k - 2 - s_2(k - \ell - 1) = 4$ . For  $n = 2$ ,  $k = 5$ ,  $\ell = 0$  the two values are 8 and 9 [Lean].

Minimizing over  $\ell$  gives  $D(n, k)$ .

**Corollary 1.3.** For all integers  $n \geq 1$  and  $k \geq 1$ ,

$$D(n, k) = \min_{0 \leq \ell < k} \Phi(n, k, \ell) = \begin{cases} 2k + n - 2 - \lfloor \log_2 k \rfloor, & k < 2^{n-1}, \\ 2k - \lfloor k/2^{n-1} \rfloor, & k \geq 2^{n-1}, \end{cases}$$

and some *k-admissible* polynomial has degree exactly  $2k - \lfloor k/2^{n-1} \rfloor$  if and only if  $k \geq 2^{n-2}$  [Lean: minimum over  $\ell$ ] [Lean: value and attainment].

Three features of these formulas stand out.

- **Blocks of  $2^n$ .** As a function of  $k - \ell$ , the excess  $\Phi - n - 2\ell$  grows by  $2^{n+1} - 2$  with each block of  $2^n$  [Lean]. The product  $F^2$  has degree at most  $2^{n+1} - 2$  and multiplicity at least  $2^n$  at every nonzero point (Lemma 3.6) [Lean], and the construction pays each block with one factor  $F^2$  and the remainder  $r$  with the truncated product  $g_{r+1}$ .
- **Where the Schwartz–Zippel bound is exact.** It is attained if and only if  $k \geq 2^{n-2}$ . For  $k < 2^{n-2}$ ,  $D(n, k)$  exceeds it by  $n - 2 - \lfloor \log_2 k \rfloor \geq 1$ , and for fixed  $k$  each further dimension costs exactly one degree. For example  $D(n, 1) = n$ ,  $D(n, 4) = n + 4$ ,  $D(n, 8) = n + 11$  and  $D(n, 15) = n + 25$  for  $n \geq \lfloor \log_2 k \rfloor + 2$  [Lean].
- **Comparison with the reals.** Over  $\mathbb{R}$ , for  $k \geq 2$  and  $n \geq 2k - 3$ , the least degree with origin order  $\ell$  is  $n + 2k - 3$  for every  $\ell \leq k - 2$ , and  $n + 2k - 2$  for  $\ell = k - 1$  [SW20, Theorems 1.4 and 1.5]. In that range  $k - \ell - 1 < 2^n$ , so over  $\mathbb{F}_2$  it is  $n + 2k - 2 - s_2(k - \ell - 1)$ : lower by  $s_2(k - \ell - 1) - 1 \geq 0$  for  $\ell \leq k - 2$ , and the same for  $\ell = k - 1$  [Lean].

## 1.2 Relation to previous work

- **Menezes [Men26].** In a recent preprint, Menezes determines, over every field  $\mathbb{F}$ , the least degree  $\delta_{\mathbb{F}}(n, k, \ell)$  of a polynomial with order at least  $k$  at the nonzero vertices of  $\{0, 1\}^n$  and order exactly  $\ell$  at the origin, for  $k \geq 2$  and  $n \geq k - 1$ . In characteristic two his formula is  $n + 2k - 2 - s_2(k - \ell - 1)$  [Men26, Corollary 1.3]. For  $n \geq k - 1$  we have  $k - \ell - 1 < 2^n$ , so Theorem 1.2 agrees with it there [Lean] and extends it, over  $\mathbb{F}_2$ , to every  $n \geq 1$  and every  $k \geq 1$ . The new range is  $n < k - 1$ , and there the formula changes form as soon as  $k - \ell - 1 \geq 2^n$ . Our upper-bound construction multiplies Menezes’ Catalan truncation, specialized to  $\mathbb{F}_2$ , by powers of  $F$ . The telescoping identity of Lemma 3.3 gives a short self-contained proof that the truncation works in every dimension. His degree formula is stated for  $n \geq k - 1$ , the range of his theorem. We read his statements, not his proofs, so we do not claim that his argument is limited to that range. We checked on 25 September 2026 that arXiv lists only version 1 of [Men26]. Our lower bound is proved independently of [Men26].
- **Bishnoi, Boyadzhyska, Das and Mészáros [BBDM23].** They determine the least size of a multiset of affine hyperplanes (and, more generally, subspaces of codimension  $d$ ) covering the nonzero points of  $\mathbb{F}_2^n$  at least  $k$  times and the origin at most  $k - 1$  times, for  $k \geq 2^{n-2}$ . With Proposition A.2, which answers their question on the polynomial version, their covers determine  $D(n, k)$  for  $k \geq 2^{n-2}$ . Corollary 1.3 completes the range  $k < 2^{n-2}$ , and Theorem 1.2 refines every value by the origin order. They refine their covers in the same way, by the number of times the origin is covered [BBDM23, Section 2 of arXiv v1]. The factor  $F$  is the product of the linear forms of the  $2^n - 1$  hyperplanes of their covers that avoid the origin.
- **Sauermann and Wigderson [SW20].** Their theorems are the real case described above. The case  $\ell = k - 1$ , with value  $n + 2k - 2$  for every  $n \geq 1$ , is their Theorem 1.5; they state it over  $\mathbb{R}$ , and Bishnoi et al. note that its proof is valid over every field [BBDM23, Section 1.3 of arXiv v1]. So the case  $\ell = k - 1$  of Theorem 1.2, where  $\Phi(n, k, k - 1) = n + 2k - 2$ , is not new [Lean]. In characteristic two they give a polynomial of degree  $n + 4$  for  $k = 4$ , below  $n + 2k - 3 = n + 5$ , showing that the real formula fails there.
- **Dvir, Kopparty, Saraf and Sudan [DKSS13].** Their multiplicity Schwartz–Zippel lemma is used only in the second proof of Corollary 1.3 (Appendix A) and for Proposition A.2.

- **Alon [Alo99] and Alon and Füredi [AF93].** The lower bound of Theorem 1.2 uses the grid-vanishing lemma of the combinatorial Nullstellensatz [Alo99, Lemma 2.1], for a grid of  $2^n$  polynomials. For  $k = 1$ , Corollary 1.3 says  $D(n, 1) = n$  for  $n \geq 2$  [Lean]; the lower bound is the case  $\mathbb{F} = \mathbb{F}_2$  of [AF93, Theorem 5].

Two web searches on 24 September 2026, for high-order vanishing on the hypercube in positive characteristic and for punctured multiplicity Nullstellensätze over  $\mathbb{F}_2$ , found no statement for the range  $n < k - 1$  other than the results above. A bounded search does not establish that none exists.

### 1.3 Notation

Throughout,  $n$  is the number of variables,  $k$  the multiplicity required at nonzero points and  $\ell < k$  the origin order. We write  $m = k - \ell - 1 = q2^n + r$  with  $0 \leq r < 2^n$ , and  $h = 2q + s_2(r)$ , so that  $\Phi = n + 2k - 2 - h$  (Lemma 3.2). Further,  $y_i = x_i^2 + x_i$  (in Section 4,  $y_i$  is an independent variable, and substituting  $y_i \mapsto x_i^2 + x_i$  recovers this meaning);  $F = \prod_{u \neq \mathbf{0}} (1 + u \cdot x)$  is the hyperplane product (Section 3);  $g_s$  is the truncated product (Definition 3.4);  $\hat{x}_K(Y) = \sum_{s < K} Y^{2^s}$  is the local inverse of  $x \mapsto x^2 + x$ , and  $w = P(\hat{x}_K(y_1), \dots, \hat{x}_K(y_n))$  (Section 4). For a polynomial  $f$ ,  $f_d$  is its homogeneous component of degree  $d$ .

### 1.4 Proof outline

- **Upper bound (Section 3).** With  $y = x^2 + x$  we have  $(1 + x) + \sum_{j < K} y^{2^j} = (1 + x)^{2^K}$  over  $\mathbb{F}_2$ . Expanding  $\prod_i (1 + x_i)^{2^K}$  this way and discarding the terms of cost at least  $s$  leaves a polynomial  $g_s$ . It still vanishes to order  $s$  off the origin, takes the value 1 at the origin, and has small degree (Theorem 3.5). The product  $F$  has multiplicity at least  $2^{n-1}$  at every nonzero point. Multiplying  $F^{2^q} g_{r+1}$  by  $y_1^\ell$  sets the origin order (Theorem 3.7).
- **Lower bound (Section 4).** Write  $P = \sum_\varepsilon x^\varepsilon A_\varepsilon(x^2 + x)$  with squarefree  $x^\varepsilon$ . A nonzero coefficient of  $y^\varepsilon$  in  $A_\varepsilon$  forces  $\deg P \geq |\varepsilon| + 2|e|$ . The multiplicity conditions determine the low-degree parts of every  $A_\varepsilon$  from one power series  $w$ , the composition of  $P$  with the additive local inverse  $\hat{x}(y) = \sum_s y^{2^s}$  of  $x^2 + x$ . So a polynomial of degree below  $\Phi$  makes certain parts of  $w \prod_{i \in T} \hat{x}(y_i)$  vanish. These parts are values of additive forms, and a recursion for repeated arguments shows that the forms vanish on the whole span  $V$  of  $y_1, \dots, y_n$ . Reducing modulo  $\prod_{v \in V} (Y - v)$  and applying the grid-vanishing lemma then isolates the lowest part  $w_\ell$ , which is nonzero.
- **Second proof of Corollary 1.3 (Appendix A).** After an invertible linear change of coordinates,  $S = P(x', 0) + P(x', 1)$  is admissible in  $n - 1$  variables, has the same origin order, and has degree at least one lower (Theorem A.6). Iterating down to  $n_0 = \lfloor \log_2 k \rfloor + 2$ , where the multiplicity Schwartz–Zippel lemma already gives degree  $2k$ , yields  $D(n, k) \geq 2k + n - n_0$ . This was the proof of version 1 of this paper.

### 1.5 Formal verification

Every mathematical claim of this paper is proved in Lean 4 with Mathlib: the theorems and lemmas, the remarks, the examples and the comparisons made in the text, together with the multiplicity Schwartz–Zippel lemma, the hyperplane covers and the parity of Catalan numbers. The proofs use only Lean’s standard axioms. Each claim carries a [Lean] link to its formal proof at a fixed

repository commit, except the grid-vanishing lemma (Lemma 4.5), which the Lean proof takes from Mathlib. Appendix B maps statements to declarations and gives the commands to check them. Not formalized are statements about the content of other papers, which we cite rather than prove: what their theorems say over other fields or in other ranges, and how their constructions relate to ours. Formal verification covers the statements under the linked definitions. It does not cover the prose, the comparison with the literature, or the fidelity of the definitions to the informal notions, although the definitions are short (Definitions 2.1 and 1.1).

**Changes from version 1.** Version 1 (24 September 2026) determined  $D(n, k)$  by the argument now in Appendix A, and gave the upper bound  $n + 2k - 2 - s_2(k - \ell - 1)$  for each origin order (Proposition 3.8), leaving its optimality open for  $n < k - 1$ . Version 2 determines  $\delta(n, k, \ell)$  in every dimension (Theorem 1.2). That bound is optimal when  $k - \ell - 1 < 2^n$ , and the new construction of Theorem 3.7 has degree below that bound otherwise [Lean]. The title, abstract and introduction are rewritten around the per-order result, Sections 4 and 5 are new, and the Lean links point to a later commit. Version 1 contains no mathematical error that this version corrects.

## 2 Multiplicity

Let  $\mathbb{F}$  be a field and  $\sigma$  a finite set of variables. Degrees are total degrees of formal polynomials in  $\mathbb{F}[x_\sigma]$ . For a multi-index  $\beta \in \mathbb{Z}_{\geq 0}^\sigma$  write  $|\beta| = \sum_i \beta_i$ .

**Definition 2.1** (Hasse multiplicity). For  $P \in \mathbb{F}[x_\sigma]$  and  $a \in \mathbb{F}^\sigma$ , write  $P(a + z) = \sum_\beta c_\beta z^\beta$ . The *multiplicity* of  $P$  at  $a$  is

$$\text{mult}_a(P) = \min\{|\beta| : c_\beta \neq 0\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\},$$

with  $\text{mult}_a(0) = \infty$  [Lean].

Dvir, Kopparty, Saraf and Sudan call the coefficient  $c_\beta$  the Hasse derivative  $P^{(\beta)}(a)$ , and this is their definition [DKSS13, Definition 2.2]. In Lean,  $\text{mult}_a(P)$  is the order, in  $\mathbb{N}_\infty$ , of the power series of the shifted polynomial  $P(a + z)$ .

**Lemma 2.2.** *Let  $P, Q \in \mathbb{F}[x_\sigma]$  and  $a \in \mathbb{F}^\sigma$ .*

1. *For  $m \in \mathbb{Z}_{\geq 0}$ ,  $\text{mult}_a(P) \geq m$  if and only if every coefficient of  $P(a + z)$  of degree less than  $m$  vanishes.*
2.  $\text{mult}_a(P + Q) \geq \min(\text{mult}_a(P), \text{mult}_a(Q))$ .
3.  $\text{mult}_a(PQ) = \text{mult}_a(P) + \text{mult}_a(Q)$ . *(The inequality  $\geq$  holds over every commutative ring, equality over every integral domain.)*
4.  $\text{mult}_a(P) = 0$  if and only if  $P(a) \neq 0$ .
5. *If  $e : \sigma \rightarrow \tau$  is a bijection and  $P^e \in \mathbb{F}[x_\tau]$  is obtained by renaming  $x_i$  to  $x_{e(i)}$ , then  $\text{mult}_b(P^e) = \text{mult}_{b \circ e}(P)$  for all  $b \in \mathbb{F}^\tau$ .*

[Lean]

*Proof.* Part 1 is the definition. Part 2 holds because a coefficient of the sum of degree below both minima is a sum of two zero coefficients. For part 3, the lowest homogeneous parts of  $P(a+z)$  and  $Q(a+z)$  multiply to the part of degree  $\text{mult}_a(P) + \text{mult}_a(Q)$  of  $(PQ)(a+z)$ ; over a field this product is nonzero when both factors are. Part 4 holds because the constant coefficient of  $P(a+z)$  is  $P(a)$ . For part 5, renaming commutes with the shift, and it maps the monomials of degree  $d$  bijectively to the monomials of degree  $d$ .  $\square$

One further fact about orders at the origin is used repeatedly. For  $f$  in a polynomial ring over  $\mathbb{F}$ , write  $f_d$  for its homogeneous component of degree  $d$ , and  $\text{ord } f = \text{mult}_0(f)$  for the least  $d$  with  $f_d \neq 0$ . Over  $\mathbb{F}_2$ , if  $z_i$  ( $i \in \sigma$ ) are elements of  $\mathbb{F}_2[y_\sigma]$  without constant term, then substituting  $x_i \mapsto z_i$  does not lower the order: a monomial of degree  $d$  goes to a product of  $d$  elements of order at least one. Hence  $\text{mult}_a(P) \leq \text{ord } P(a+z)$  for every such substitution  $z$  [Lean].

### 3 The upper bound

From now on the field is  $\mathbb{F}_2$  unless stated otherwise.

#### 3.1 Binary digit sums

Let  $s_2(m)$  be the sum of the binary digits of  $m \geq 0$ .

**Lemma 3.1.** *For all  $a, b, j, J \geq 0$ :*

1.  $s_2(a+b) \leq s_2(a) + s_2(b)$  and  $s_2(2^j) = 1$ . Hence a sum of  $r$  powers of two has digit sum at most  $r$ .
2.  $s_2(a) \leq a$ .
3. If  $a \leq b$ , then  $2a + s_2(b) \leq 2b + s_2(a)$ . That is,  $a \mapsto 2a - s_2(a)$  is monotone.
4.  $s_2(2^J - 1) = J$ .

[Lean: parts 1–3] [Lean: part 4]

*Proof.* All four parts use the recursion  $s_2(m) = (m \bmod 2) + s_2(\lfloor m/2 \rfloor)$ .

1. The first inequality is proved by strong induction on  $a+b$ , using  $\lfloor (a+b)/2 \rfloor = \lfloor a/2 \rfloor + \lfloor b/2 \rfloor + c$ , where  $c = \lfloor ((a \bmod 2) + (b \bmod 2))/2 \rfloor$  is the carry. The value  $s_2(2^j) = 1$  follows from the recursion by induction on  $j$ .
2. This is standard; Mathlib proves it for every base.
3. Induct on  $b-a$ , using  $s_2(b+1) \leq s_2(b) + 1$ , which is part 1 with second summand 1.
4.  $2^{J+1} - 1$  is odd with  $\lfloor (2^{J+1} - 1)/2 \rfloor = 2^J - 1$ .  $\square$

**Lemma 3.2** (Legendre form). *Let  $n \geq 0$ ,  $q \geq 0$  and  $0 \leq r < 2^n$ , and put  $m = q2^n + r$ . Then*

$$\sum_{j=0}^{n-1} \left\lfloor \frac{m}{2^j} \right\rfloor + 2q + s_2(r) = 2m.$$

Hence, for  $n \geq 1$ ,  $0 \leq \ell < k$  and  $k - \ell - 1 = q2^n + r$  with  $0 \leq r < 2^n$ ,  $\Phi(n, k, \ell) = n + 2k - 2 - 2q - s_2(r)$ , and  $\Phi(n, k, \ell) = n + 2k - 2 - s_2(k - \ell - 1)$  when  $k - \ell - 1 < 2^n$ . Moreover  $m$  is the sum of  $h = 2q + s_2(r)$  powers of two, each at most  $2^{n-1}$  when  $n \geq 1$ :  $2q$  copies of  $2^{n-1}$  and the binary digits of  $r$  [Lean] [Lean: form of  $\Phi$ ].

*Proof.* For  $j < n$ ,  $\lfloor m/2^j \rfloor = q2^{n-j} + \lfloor r/2^j \rfloor$ , and  $\sum_{j < n} q2^{n-j} = 2q2^n - 2q$ . Since  $r < 2^n$ ,  $\sum_{j < n} \lfloor r/2^j \rfloor = \sum_{j \geq 0} \lfloor r/2^j \rfloor$ , which is  $2r - s_2(r)$ . This is Legendre's classical identity  $\sum_{j \geq 1} \lfloor r/2^j \rfloor = r - s_2(r)$ , plus the term  $j = 0$ ; by induction, for  $r = 2r' + b$  with  $b \in \{0, 1\}$  the sum is  $r + (2r' - s_2(r')) = 2r - s_2(r)$ . Adding gives  $2m - 2q - s_2(r)$ . The formula for  $\Phi$  follows with  $m = k - \ell - 1$ , since  $n + 2\ell + 2m = n + 2k - 2$ . For the decomposition,  $2q \cdot 2^{n-1} = q2^n$ , and the binary digits of  $r < 2^n$  are powers  $2^j$  with  $j \leq n - 1$ .  $\square$

### 3.2 The truncated product

Put  $y_i = x_i^2 + x_i = x_i(1 + x_i)$ .

**Lemma 3.3.** *In every commutative ring of characteristic two and for every  $t$  and  $K \geq 0$ ,  $\sum_{j < K} (t^2 + t)^{2^j} = t^{2^K} + t$ . In particular, over  $\mathbb{F}_2$ ,*

$$(1 + x) + \sum_{j < K} y^{2^j} = (1 + x)^{2^K}, \quad y = x^2 + x$$

[Lean].

*Proof.* By Frobenius,  $(t^2 + t)^{2^j} = t^{2^{j+1}} + t^{2^j}$ , so the sum telescopes. For the second form take  $t = 1 + x$ , for which  $t^2 + t = x^2 + x$ .  $\square$

**Definition 3.4** (Truncated product). For  $s \geq 1$ , let  $g_s \in \mathbb{F}_2[x_\sigma]$  be the sum, over all subsets  $S \subseteq \sigma$  and all assignments  $i \mapsto a_i$  of powers of two to  $i \in S$  with  $\sum_{i \in S} a_i \leq s - 1$ , of

$$\prod_{i \in S} y_i^{a_i} \prod_{i \notin S} (1 + x_i)$$

[Lean].

The Catalan number  $C_{a-1}$  is odd exactly when  $a$  is a power of two; more precisely  $v_2(C_{a-1}) = s_2(a) - 1$  [AK73], [DS06, Theorem 2.1 of the arXiv version] [Lean]. So  $g_s$  is the specialization to  $\mathbb{F}_2$  of the Catalan truncation of Menezes [Men26, Section 3]. In Lean the pair  $(S, a)$  is encoded by a map  $c : \sigma \rightarrow \{\perp, 0, \dots, s - 1\}$ :  $c_i = \perp$  means  $i \notin S$ , and  $c_i = j$  means  $a_i = 2^j$ . The sum runs over maps of cost  $\sum_{c_i \neq \perp} 2^{c_i} \leq s - 1$ . The bound  $j < s$  is automatic, because  $2^j \leq s - 1 < 2^s$ . (The Lean statements hold for every  $s \geq 0$  with natural-number subtraction; only  $s \geq 1$  is used.)

**Theorem 3.5.** *For every finite  $\sigma$  with  $|\sigma| = n$  and every  $s \geq 1$ :*

1.  $\text{mult}_a(g_s) \geq s$  for every nonzero  $a \in \mathbb{F}_2^\sigma$ ;
2.  $g_s(\mathbf{0}) = 1$ ;
3.  $\deg g_s \leq n + 2(s - 1) - s_2(s - 1)$ .

[Lean]

*Proof. Expansion.* By Lemma 3.3 with  $K = s$ ,

$$\prod_{i \in \sigma} (1 + x_i)^{2^s} = \prod_{i \in \sigma} \left( (1 + x_i) + \sum_{j < s} y_i^{2^j} \right).$$

Expanding the right-hand side gives the terms of Definition 3.4 of every cost. Hence the left-hand side equals  $g_s + R$ , where  $R$  collects the terms of cost at least  $s$ . This includes every term that uses a power  $2^j > s - 1$ .

*Multiplicity (1).*

- Since  $u^2 + u = 0$  on  $\mathbb{F}_2$ , each  $y_i$  vanishes at every point of  $\mathbb{F}_2^\sigma$ . Lemma 2.2(3) then gives each term of  $R$  multiplicity at least its cost, so at least  $s$  everywhere. By Lemma 2.2(2), so does  $R$ .
- At a nonzero  $a$ , choose  $i_0$  with  $a_{i_0} = 1$ . Then  $(1 + x_{i_0})^{2^s}$  has multiplicity at least  $2^s \geq s$  at  $a$ , and so does the whole product.

In characteristic two,  $g_s$  equals the product plus  $R$ , so  $\text{mult}_a(g_s) \geq s$ .

*Value (2).* Every term with  $S \neq \emptyset$  vanishes at  $\mathbf{0}$ , and the term with  $S = \emptyset$  is  $\prod(1 + 0) = 1$ .

*Degree (3).* The term of  $(S, a)$  has degree at most  $(n - |S|) + 2w$ , where  $w = \sum_{i \in S} a_i \leq s - 1$ . By Lemma 3.1(1),  $|S| \geq s_2(w)$ . By Lemma 3.1(3),  $2w - s_2(w) \leq 2(s - 1) - s_2(s - 1)$ .  $\square$

### 3.3 The hyperplane product

Let  $F = \prod_{u \in \mathbb{F}_2^n \setminus \{\mathbf{0}\}} (1 + u \cdot x)$ , where  $u \cdot x = \sum_i u_i x_i$ . Its  $2^n - 1$  factors [Lean] define the hyperplanes  $u \cdot x = 1$ , which avoid the origin.

**Lemma 3.6.** *For  $n \geq 1$ :  $\text{mult}_a(F) \geq 2^{n-1}$  for every nonzero  $a \in \mathbb{F}_2^n$ ,  $F(\mathbf{0}) = 1$ , and  $\deg F \leq 2^n - 1$  [Lean].*

*Proof.* Fix  $a \neq \mathbf{0}$  and  $i$  with  $a_i = 1$ . Flipping  $u_i$  is an involution of  $\mathbb{F}_2^n$  that changes  $u \cdot a$  by one, so exactly  $2^{n-1}$  vectors  $u$  satisfy  $u \cdot a = 1$ . All of them are nonzero, so they index factors of  $F$  that vanish at  $a$ . Lemma 2.2(3,4) gives the first claim. Each factor takes the value 1 at  $\mathbf{0}$  and has degree at most one.  $\square$

These are the facts behind the hyperplane covers of [BBDM23] (Theorem A.8).

### 3.4 The construction

**Theorem 3.7.** *Let  $\sigma$  be finite with  $|\sigma| = n \geq 1$ , fix  $x_1 \in \sigma$ , let  $0 \leq \ell < k$ , and write  $k - \ell - 1 = q2^n + r$  with  $0 \leq r < 2^n$ . The polynomial*

$$P = y_1^\ell F^{2^q} g_{r+1}$$

*satisfies:*

1.  $\text{mult}_a(P) \geq k$  for every nonzero  $a$ ;
2.  $\text{mult}_{\mathbf{0}}(P) = \ell$ ;
3.  $\deg P \leq \Phi(n, k, \ell)$ .

Hence  $\delta(n, k, \ell) \leq \Phi(n, k, \ell)$  [Lean].

- Proof.* 1. Since  $u^2 + u = 0$  on  $\mathbb{F}_2$ ,  $y_1$  vanishes at every point, so  $\text{mult}_a(y_1^\ell) \geq \ell$ . With Lemma 3.6, Theorem 3.5(1) and Lemma 2.2(3), at  $a \neq \mathbf{0}$  we get  $\text{mult}_a(P) \geq \ell + 2q \cdot 2^{n-1} + (r+1) = \ell + (k - \ell - 1) + 1 = k$ .
2.  $y_1$  has origin order exactly one: its coefficient of  $x_1$  is 1. Also  $F(\mathbf{0}) = g_{r+1}(\mathbf{0}) = 1$ , so these factors have origin order zero. Multiplicities add over  $\mathbb{F}_2$  (Lemma 2.2(3)).
3.  $\deg y_1^\ell \leq 2\ell$ ,  $\deg F^{2q} \leq 2q(2^n - 1)$ , and  $\deg g_{r+1} \leq n + 2r - s_2(r)$  by Theorem 3.5(3). The sum is  $n + 2\ell + 2(q2^n + r) - 2q - s_2(r) = n + 2k - 2 - 2q - s_2(r)$ , which is  $\Phi(n, k, \ell)$  by Lemma 3.2.  $\square$

For  $k - \ell - 1 < 2^n$  we have  $q = 0$ , and  $P = y_1^\ell g_{k-\ell}$  is Menezes' polynomial over  $\mathbb{F}_2$ . It works in every dimension, with the digit-sum bound, also when  $k - \ell - 1 \geq 2^n$ , where Theorem 3.7 has degree below that bound [Lean].

**Proposition 3.8.** *Let  $|\sigma| = n \geq 1$ ,  $x_1 \in \sigma$  and  $0 \leq \ell < k$ . Then  $P = y_1^\ell g_{k-\ell}$  has  $\text{mult}_a(P) \geq k$  for every nonzero  $a$ ,  $\text{mult}_0(P) = \ell$ , and  $\deg P \leq n + 2k - 2 - s_2(k - \ell - 1)$  [Lean].*

*Proof.* As in the proof of Theorem 3.7, with  $g_{k-\ell}$  in place of  $F^{2q}g_{r+1}$ : the multiplicity off the origin is at least  $\ell + (k - \ell)$ , the origin order is  $\ell$ , and the degree is at most  $2\ell + n + 2(k - \ell - 1) - s_2(k - \ell - 1)$  by Theorem 3.5(3).  $\square$

Menezes proves the degree  $n + 2k - 2 - s_2(k - \ell - 1)$  optimal when  $n \geq k - 1$  [Men26, Corollary 1.3]. Section 4 proves Theorem 3.7 optimal in every dimension; the construction has degree exactly  $\Phi(n, k, \ell)$  [Lean].

## 4 The lower bound at every origin order

Throughout this section  $\sigma$  is finite with  $|\sigma| = n \geq 1$ . Besides  $\mathbb{F}_2[x_\sigma]$  we use a second polynomial ring  $\mathbb{F}_2[y_\sigma]$  in independent variables  $y_i$ . For  $A \in \mathbb{F}_2[y_\sigma]$ ,  $A(x^2 + x)$  denotes the substitution  $y_i \mapsto x_i^2 + x_i$ . For  $\varepsilon \subseteq \sigma$  put  $x^\varepsilon = \prod_{i \in \varepsilon} x_i$ , and for an exponent vector  $e$  put  $|e| = \sum_i e_i$ . Recall that  $f_d$  is the homogeneous component of degree  $d$  of  $f$ , and put  $f_d = 0$  for  $d < 0$ . For  $K \geq 0$  let

$$\hat{x}_K(Y) = \sum_{s < K} Y^{2^s}.$$

Over  $\mathbb{F}_2$ ,  $\hat{x}_K(Y)^2 + \hat{x}_K(Y) = Y + Y^{2^K}$ , by the telescoping of Lemma 3.3 [Lean]. So  $\hat{x}_K(y)$  inverts  $x \mapsto x^2 + x$  up to order  $2^K$ ; it is the truncation below degree  $2^K$  of the classical additive power series  $\sum_{s \geq 0} Y^{2^s}$ , which solves the Artin–Schreier equation  $X^2 + X = Y$  in  $\mathbb{F}_2[[Y]]$  [Lean].

**Plan.** Suppose  $\deg P < \Phi$ . Lemmas 4.1 and 4.2 turn the degree bound and the multiplicities into the vanishing of certain homogeneous parts of  $w \prod_{i \in T} \hat{x}_K(y_i)$ , and Lemma 4.3 shows that  $w$  starts in degree exactly  $\ell$ . These parts are values, at the generators  $y_i$ , of additive forms (Lemma 4.4), and a recursion carries the vanishing from tuples of distinct generators to all tuples, hence to the span  $V$ . Vanishing on  $V$  is a statement about values, not coefficients. The reduction modulo  $\prod_{v \in V} (Y - v)$  (Lemma 4.6) turns it into a statement about coefficients, and one coefficient is  $w_\ell$ .

## 4.1 The expansion

**Lemma 4.1** (Expansion). *Every  $P \in \mathbb{F}_2[x_\sigma]$  can be written as*

$$P = \sum_{\varepsilon \subseteq \sigma} x^\varepsilon A_\varepsilon(x^2 + x), \quad A_\varepsilon \in \mathbb{F}_2[y_\sigma].$$

*For every such expansion, if the monomial  $y^e$  occurs in  $A_\varepsilon$ , then  $|\varepsilon| + 2|e| \leq \deg P$  [Lean].*

*Proof.* For existence it suffices to expand a monomial, and by multiplicativity a power  $x_i^b$ . By induction on  $b$ ,  $x_i^b$  is a sum of terms  $x_i^\eta y_i^c$  with  $\eta \in \{0, 1\}$ : this holds for  $b \leq 1$ , and  $x_i^b = x_i^{b-2}(x_i^2 + x_i) + x_i^{b-1}$ .

For the degree bound, let  $W$  be the largest value of  $|\varepsilon| + 2|e|$  over the pairs  $(\varepsilon, e)$  with  $y^e$  in  $A_\varepsilon$ , with coefficient  $c_{\varepsilon, e}$ . The polynomial  $x^\varepsilon(x^2 + x)^e$  has degree  $|\varepsilon| + 2|e|$ , and its homogeneous component of that degree is the single monomial  $x^{\mathbf{1}_\varepsilon + 2e}$ . The map  $(\varepsilon, e) \mapsto \mathbf{1}_\varepsilon + 2e$  is injective, since  $\varepsilon$  is the set of odd coordinates. So the component of degree  $W$  of  $P$  is  $\sum c_{\varepsilon, e} x^{\mathbf{1}_\varepsilon + 2e}$  over the pairs of weight  $W$ , a sum of distinct monomials, which is nonzero. Hence  $\deg P \geq W$ .  $\square$

## 4.2 The multiplicity conditions

**Lemma 4.2** (Congruence). *Let  $k \geq 1$  and  $K$  with  $2^K \geq k$ . Let  $P \in \mathbb{F}_2[x_\sigma]$  have  $\text{mult}_a(P) \geq k$  for every nonzero  $a \in \mathbb{F}_2^\sigma$ , and let  $P = \sum_\varepsilon x^\varepsilon A_\varepsilon(x^2 + x)$  be any expansion. Put  $\hat{x}_i = \hat{x}_K(y_i)$  and  $w = P(\hat{x}) \in \mathbb{F}_2[y_\sigma]$ . Then for every  $\varepsilon \subseteq \sigma$  and every  $d < k$ ,*

$$(A_\varepsilon)_d = \left( w \prod_{i \notin \varepsilon} (1 + \hat{x}_i) \right)_d$$

[Lean].

*Proof.* Write  $\equiv$  for equality of all homogeneous components of degree below  $k$ .

- *The substitution.* For  $\delta \in \mathbb{F}_2^\sigma$  substitute  $x_i \mapsto \hat{x}_i + \delta_i$ . Since  $\delta_i^2 = \delta_i$  and the cross term vanishes in characteristic two,  $x_i^2 + x_i$  goes to  $\hat{x}_i^2 + \hat{x}_i = y_i + y_i^{2^K} =: \hat{y}_i$  for both values of  $\delta_i$ . Hence

$$P(\hat{x} + \delta) = \sum_{\varepsilon'} \prod_{i \in \varepsilon'} (\hat{x}_i + \delta_i) A_{\varepsilon'}(\hat{y}).$$

- *Nonzero shifts.* For  $\delta \neq \mathbf{0}$ ,  $P(\hat{x} + \delta)$  is  $P(\delta + z)$  at  $z_i = \hat{x}_i$ , which have no constant term. Since  $\text{mult}_\delta(P) \geq k$ ,  $P(\hat{x} + \delta)$  has order at least  $k$  (Section 2), so  $P(\hat{x} + \delta) \equiv 0$ .
- *Inversion.* For  $c \in \mathbb{F}_2$  put  $N_i(c) = 1$  if  $i \in \varepsilon$ , and otherwise  $N_i(0) = 1 + \hat{x}_i$ ,  $N_i(1) = \hat{x}_i$ . A check of the four cases gives

$$\sum_{c \in \mathbb{F}_2} N_i(c) (\hat{x}_i + c)^{[i \in \varepsilon']} = [i \in \varepsilon \Leftrightarrow i \in \varepsilon'].$$

For instance, for  $i \notin \varepsilon$  and  $i \in \varepsilon'$  the sum is  $(1 + \hat{x}_i)\hat{x}_i + \hat{x}_i(\hat{x}_i + 1) = 0$ . Multiplying over  $i$  and summing over  $\delta \in \mathbb{F}_2^\sigma$  gives  $\sum_\delta \prod_i N_i(\delta_i) \prod_{i \in \varepsilon'} (\hat{x}_i + \delta_i) = [\varepsilon' = \varepsilon]$ , and therefore

$$A_\varepsilon(\hat{y}) = \sum_\delta \prod_i N_i(\delta_i) P(\hat{x} + \delta).$$

- *Conclusion.* Modulo  $\equiv$  only  $\delta = \mathbf{0}$  survives, and it contributes  $\prod_{i \notin \varepsilon} (1 + \hat{x}_i) w$ . Finally  $\hat{y}_i - y_i = y_i^{2^K}$  has order  $2^K \geq k$ , so  $A_\varepsilon(\hat{y}) \equiv A_\varepsilon(y)$ .  $\square$

**Lemma 4.3** (The lowest part). *In Lemma 4.2, if moreover  $\text{mult}_0(P) = \ell < k$ , then  $w_d = 0$  for  $d < \ell$  and  $w_\ell \neq 0$  [Lean].*

*Proof.* Since the  $\hat{x}_i$  have no constant term,  $\text{ord } w \geq \ell$  (Section 2). Substitute  $y_i \mapsto x_i^2 + x_i$  in  $w$ . By Lemma 3.3 with  $t = x_i$ ,  $\hat{x}_K(x_i^2 + x_i) = x_i + x_i^{2^K}$ , so the result is  $P(x + x^{2^K})$ . The difference  $P(x + x^{2^K}) - P(x)$  lies in the ideal generated by the  $x_i^{2^K}$ , so it has order at least  $2^K \geq k > \ell$ , and  $P(x + x^{2^K})$  has order exactly  $\ell$ . If  $w_\ell$  were zero, then  $\text{ord } w > \ell$ , and since each  $x_i^2 + x_i$  has order one,  $P(x + x^{2^K})$  would have order above  $\ell$ .  $\square$

### 4.3 Additive forms

Fix  $k \geq 1$  and  $w \in \mathbb{F}_2[y_\sigma]$ . For  $t \geq 0$ ,  $j \geq 0$  and  $Y_1, \dots, Y_j \in \mathbb{F}_2[y_\sigma]$  put

$$E_j^{(t)}(Y_1, \dots, Y_j) = \sum_{\tau \in \{0, \dots, k-1\}^j} Y_1^{2^{\tau_1}} \cdots Y_j^{2^{\tau_j}} w_{k-1-t-\sum_a 2^{\tau_a}}.$$

**Lemma 4.4** (Forms). 1.  $E_j^{(t)}$  is symmetric, and additive in each argument.

2. For  $t < k$  and  $i_1, \dots, i_j \in \sigma$ ,  $E_j^{(t)}(y_{i_1}, \dots, y_{i_j}) = (w \prod_a \hat{x}_k(y_{i_a}))_{k-1-t}$ .

3. For  $j \geq 0$  and all  $Y, Z_1, \dots, Z_j$ ,

$$E_{j+2}^{(t)}(Y, Y, Z) = E_{j+1}^{(t)}(Y, Z) + Y E_j^{(t+1)}(Z).$$

[Lean]

*Proof.* 1. Permuting the arguments permutes the index set. Additivity holds because  $Y \mapsto Y^{2^s}$  is additive in characteristic two.

2. Expand  $\prod_a \hat{x}_k(y_{i_a}) = \sum_\tau \prod_a y_{i_a}^{2^{\tau_a}}$  over  $\tau \in \{0, \dots, k-1\}^j$ . Each product is homogeneous of degree  $\sum_a 2^{\tau_a}$ , so the component of degree  $k-1-t$  of  $w$  times it is  $\prod_a y_{i_a}^{2^{\tau_a}} w_{k-1-t-\sum_a 2^{\tau_a}}$ .

3. In  $E_{j+2}^{(t)}(Y, Y, Z)$  the terms with  $\tau_1 \neq \tau_2$  cancel in pairs under the exchange of  $\tau_1$  and  $\tau_2$ . The terms with  $\tau_1 = \tau_2 = s$  give  $Y^{2^{s+1}} w_{k-1-t-2^{s+1}-\Sigma}$ , where  $\Sigma$  is the contribution of  $Z$ . In  $E_{j+1}^{(t)}(Y, Z)$ , the terms with  $Y$ -exponent  $2^{s+1}$ ,  $0 \leq s < k-1$ , are the same terms. The remaining diagonal term  $s = k-1$  is zero, because  $2^k > k-1-t$  makes the index of  $w$  negative. The terms of  $E_{j+1}^{(t)}(Y, Z)$  with  $Y$ -exponent 1 are  $Y w_{k-1-(t+1)-\Sigma}$ , which sum to  $Y E_j^{(t+1)}(Z)$ . Signs do not matter in characteristic two.  $\square$

### 4.4 Reduction on a grid

**Lemma 4.5** (Grid vanishing [Alo99, Lemma 2.1]). *Let  $D$  be an integral domain,  $V \subseteq D$  finite, and  $R \in D[Y_1, \dots, Y_h]$  of degree less than  $|V|$  in each variable. If  $R$  vanishes on  $V^h$ , then  $R = 0$ .*

*Proof.* Induct on  $h$ . Write  $R = \sum_{e < |V|} R_e(Y_1, \dots, Y_{h-1}) Y_h^e$ . For fixed  $v' \in V^{h-1}$ , the polynomial  $R(v', Y_h)$  has degree less than  $|V|$  and  $|V|$  roots, so it is zero, and  $R_e(v') = 0$  for every  $e$ . By induction every  $R_e$  is zero.  $\square$

Mathlib proves this lemma as part of the combinatorial Nullstellensatz, and the Lean proof uses that version.

**Lemma 4.6** (Grid reduction). *Let  $D$  be an integral domain,  $V \subseteq D$  finite and nonempty,  $L_V = \prod_{v \in V} (Y - v)$ , and  $\rho_N$  the remainder of  $Y^N$  modulo  $L_V$ . Let  $I$  be finite,  $c_i \in D$  and  $d_{i,a} \geq 0$  for  $i \in I$  and  $a = 1, \dots, h$ . If*

$$\sum_{i \in I} c_i \prod_{a=1}^h v_a^{d_{i,a}} = 0 \quad \text{for all } v \in V^h,$$

*then for all exponents  $e_1, \dots, e_h < |V|$ ,*

$$\sum_{i \in I} c_i \prod_{a=1}^h [Y^{e_a}] \rho_{d_{i,a}} = 0$$

[Lean].

*Proof.* Let  $R = \sum_i c_i \prod_a \rho_{d_{i,a}}(Y_a)$ . It has degree less than  $|V|$  in each variable, because  $\deg \rho_N < |V|$ . Since  $L_V$  divides  $Y^N - \rho_N$  and vanishes on  $V$ ,  $\rho_N(v) = v^N$  for  $v \in V$ , so  $R$  vanishes on  $V^h$ . By Lemma 4.5,  $R = 0$ . Its coefficient of  $\prod_a Y_a^{e_a}$  is the displayed sum.  $\square$

## 4.5 The lower bound

**Theorem 4.7.** *Let  $n \geq 1$  and  $0 \leq \ell < k$ . Every  $P \in \mathbb{F}_2[x_\sigma]$  with  $\text{mult}_a(P) \geq k$  for all nonzero  $a$  and  $\text{mult}_0(P) = \ell$  satisfies  $\deg P \geq \Phi(n, k, \ell)$  [Lean].*

*Proof.* Let  $m = k - \ell - 1 = q2^n + r$  with  $0 \leq r < 2^n$ , and  $h = 2q + s_2(r)$ . By Lemma 3.2,  $\Phi(n, k, \ell) = n + 2(k - 1) - h$ , and  $m = \sum_{a=1}^h 2^{s_a}$  with  $s_a \leq n - 1$  for all  $a$ . Since every part is at least one,  $h \leq m \leq k - 1$ , and  $2^{s_a} \leq m$  gives  $s_a < k$ .

Suppose that  $\deg P \leq \Phi(n, k, \ell) - 1$ . Take  $K = k$ , so  $2^K \geq k$ , an expansion of  $P$  (Lemma 4.1), and  $w = P(\hat{x})$  as in Lemma 4.2. Form  $E_j^{(t)}$  from this  $w$ .

*Step 1: the conditions  $(C_t)$ .* For  $0 \leq t < k$  and  $T \subseteq \sigma$  with  $|T| + 2t \leq h$ ,

$$\left( w \prod_{i \in T} \hat{x}_i \right)_{k-1-t} = 0.$$

First let  $S \subseteq \sigma$  with  $|S| + 2t \leq h$ , and  $\varepsilon = \sigma \setminus S$ . A monomial  $y^e$  with  $|e| = k - 1 - t$  in  $A_\varepsilon$  would have  $|\varepsilon| + 2|e| = n - |S| + 2(k - 1) - 2t \geq n + 2(k - 1) - h > \deg P$ , contradicting Lemma 4.1. So  $(A_\varepsilon)_{k-1-t} = 0$ , and Lemma 4.2 gives  $(w \prod_{i \in S} (1 + \hat{x}_i))_{k-1-t} = 0$ . Expanding  $\prod_{i \in S} (1 + \hat{x}_i) = \sum_{T \subseteq S} \prod_{i \in T} \hat{x}_i$ , and using that every  $T \subseteq S$  also satisfies  $|T| + 2t \leq h$ , induction on  $|S|$  gives the claim for  $T = S$ .

*Step 2: vanishing on the span.* We show, by induction on  $j$  and for all  $t$  at once, that  $E_j^{(t)}(y_{i_1}, \dots, y_{i_j}) = 0$  for all  $i_1, \dots, i_j \in \sigma$  whenever  $j + 2t \leq h$ . Then  $t \leq h/2 < k$ .

- If the indices are distinct, this is Step 1 with  $T = \{i_1, \dots, i_j\}$ , by Lemma 4.4(2). This covers  $j \leq 1$ .
- Otherwise, by symmetry, the tuple is  $(y_i, y_i, Z)$ , and by Lemma 4.4(3) the value is  $E_{j-1}^{(t)}(y_i, Z) + y_i E_{j-2}^{(t+1)}(Z)$ . Both terms vanish by induction, since  $(j - 1) + 2t \leq h$  and  $(j - 2) + 2(t + 1) \leq h$ .

Let  $V = \{\sum_{i \in U} y_i : U \subseteq \sigma\}$ , the  $\mathbb{F}_2$ -span of the variables, with  $|V| = 2^n$ . By additivity (Lemma 4.4(1)),  $E_h^{(0)}$  vanishes on  $V^h$ .

*Step 3: reduction.* Apply Lemma 4.6 with  $D = \mathbb{F}_2[y_\sigma]$ , this  $V$ ,  $I = \{0, \dots, k-1\}^h$ ,  $c_\tau = w_{k-1-\sum_a 2^{\tau a}}$ ,  $d_{\tau,a} = 2^{\tau a}$ , and  $e_a = 2^{s_a} < 2^n$ . It gives

$$\sum_{\tau} w_{k-1-\sum_a 2^{\tau a}} \prod_a [Y^{2^{s_a}}] \rho_{2^{\tau a}} = 0.$$

Since  $L_V$  is monic of degree  $2^n$ ,  $\rho_N = Y^N$  for  $N < 2^n$ .

- If  $\tau_a < s_a$  for some  $a$ , then  $\rho_{2^{\tau a}} = Y^{2^{\tau a}}$  has no  $Y^{2^{s_a}}$  term, and the term vanishes.
- If  $\tau_a \geq s_a$  for all  $a$  and  $\tau \neq s$ , then  $\sum_a 2^{\tau a} > m$ , so the index  $k-1-\sum_a 2^{\tau a}$  is below  $\ell$ , and  $w$  has no such component by Lemma 4.3.
- For  $\tau = s$ ,  $\rho_{2^{s_a}} = Y^{2^{s_a}}$ , and the term is  $w_{k-1-m} = w_\ell$ .

So  $w_\ell = 0$ , contradicting Lemma 4.3.  $\square$

In the construction, each block of  $2^n$  in  $m$  is paid by a factor  $F^2$ ; correspondingly, each block contributes two parts  $2^{n-1}$  to the decomposition of  $m$  used in this proof.

## 5 Proof of the main results

*Proof of Theorem 1.2.* Theorem 3.7 gives a polynomial with the required multiplicities and degree at most  $\Phi(n, k, \ell)$ , and Theorem 4.7 shows that every such polynomial has degree at least  $\Phi(n, k, \ell)$ . So the constructed polynomial has degree exactly  $\Phi(n, k, \ell)$ .  $\square$

**Lemma 5.1** (The minimum over the origin order). *For  $n \geq 1$  and  $k \geq 1$ ,  $\min_{0 \leq \ell < k} \Phi(n, k, \ell)$  equals  $2k + n - 2 - \lfloor \log_2 k \rfloor$  for  $k < 2^{n-1}$ , and  $2k - \lfloor k/2^{n-1} \rfloor$  for  $k \geq 2^{n-1}$  [Lean: this proof] [Lean: via Appendix A].*

*Proof.* Write  $m = k - \ell - 1 = q2^n + r$  as before and  $h(m) = 2q + s_2(r)$ . By Lemma 3.2,  $\Phi = n + 2k - 2 - h(m)$ , so the minimum is  $n + 2k - 2 - \max_{0 \leq m \leq k-1} h(m)$ . Write  $k-1 = Q2^n + R$  with  $0 \leq R < 2^n$ . Let  $\lambda = \lfloor \log_2(R+1) \rfloor$ . The largest digit sum among  $0, \dots, R$  is  $\lambda$ : it is attained at  $2^\lambda - 1 \leq R$ , and every number up to  $R$  is below  $2^{\lambda+1} - 1$ , so it has at most  $\lambda$  binary ones. With  $q = Q$  the best value of  $h$  is  $2Q + \lambda$ . With  $q \leq Q-1$ , possible only if  $Q \geq 1$ , it is at most  $2Q - 2 + n$ , attained at  $m = Q2^n - 1$ .

- If  $Q = 0$ , that is  $k \leq 2^n$ , the minimum is  $n + 2k - 2 - \lfloor \log_2 k \rfloor$ . This is the first case when  $k < 2^{n-1}$ . For  $2^{n-1} \leq k < 2^n$  it is  $2k - 1$ , and for  $k = 2^n$  it is  $2k - 2$ ; both equal  $2k - \lfloor k/2^{n-1} \rfloor$ .
- If  $Q \geq 1$ , then  $k = Q2^n + (R+1)$  with  $1 \leq R+1 \leq 2^n$ , and  $\lfloor k/2^{n-1} \rfloor = 2Q + \lfloor (R+1)/2^{n-1} \rfloor$ . Moreover  $\max(\lambda, n-2) = n-2 + \lfloor (R+1)/2^{n-1} \rfloor$  in each of the three ranges  $R+1 < 2^{n-1}$ ,  $2^{n-1} \leq R+1 < 2^n$  and  $R+1 = 2^n$  (for  $n=1$  the first is empty). So the minimum is  $n + 2k - 2 - 2Q - (n-2) - \lfloor (R+1)/2^{n-1} \rfloor = 2k - \lfloor k/2^{n-1} \rfloor$ .  $\square$

The first Lean proof of Lemma 5.1 follows this digit argument. A second one derives the minimum from Theorem 1.2 and the value of  $D(n, k)$  proved in Appendix A: every  $\Phi(n, k, \ell)$  is the degree of a  $k$ -admissible polynomial, and the origin order of a  $k$ -admissible polynomial of least degree attains the minimum.

*Proof of Corollary 1.3.* Since  $D(n, k) = \min_{\ell} \delta(n, k, \ell)$ , Theorem 1.2 and Lemma 5.1 give the value. By Proposition A.2, or directly from the value, every  $k$ -admissible polynomial has degree at least  $2k - \lfloor k/2^{n-1} \rfloor$ . So a polynomial of exactly this degree exists if and only if  $D(n, k)$  equals it. Compare the two in three cases.

- If  $k \geq 2^{n-1}$ , they are equal.
- If  $2^{n-2} \leq k < 2^{n-1}$ , then  $\lfloor \log_2 k \rfloor = n - 2$  and  $\lfloor k/2^{n-1} \rfloor = 0$ , so both equal  $2k$ .
- If  $k < 2^{n-2}$ , then  $n \geq 3$ ,  $\lfloor \log_2 k \rfloor \leq n - 3$  and  $\lfloor k/2^{n-1} \rfloor = 0$ , so  $D(n, k) \geq 2k + 1$ .  $\square$

**Corollary 5.2.** For  $n \geq 1$  and  $1 \leq k < 3 \cdot 2^{n-1}$ ,  $D(n, k) = 2k + n - 2 - \lfloor \log_2 k \rfloor$  [Lean].

*Proof.* For  $k < 2^{n-1}$  this is Corollary 1.3. Otherwise compare with  $2k - \lfloor k/2^{n-1} \rfloor$ .

- If  $2^{n-1} \leq k < 2^n$ , then  $\lfloor \log_2 k \rfloor = n - 1$  and  $\lfloor k/2^{n-1} \rfloor = 1$ , so both expressions equal  $2k - 1$ .
- If  $2^n \leq k < 3 \cdot 2^{n-1}$ , then  $\lfloor \log_2 k \rfloor = n$  and  $\lfloor k/2^{n-1} \rfloor = 2$ , so both equal  $2k - 2$ .  $\square$

At  $k = 3 \cdot 2^{n-1}$  the logarithmic expression is  $2k - 2$ , while  $D(n, k) = 2k - 3$  [Lean].

*Remark 5.3* (Which origin orders are extremal). The orders  $\ell$  attaining  $D(n, k)$  are those for which  $k - \ell - 1 = q2^n + r$  maximizes  $2q + s_2(r)$ . For  $k \leq 2^n$  they are the  $\ell$  with  $s_2(k - \ell - 1) = \lfloor \log_2 k \rfloor$ , for instance  $\ell = k - 2^{\lfloor \log_2 k \rfloor}$  [Lean]. Exact rank computations of  $\delta(n, k, \ell)$  recorded in the [research notebook](#) agree with  $\Phi$ : all 696 values for  $n \leq 5$ ,  $k \leq 16$  and for  $(n, k) = (6, 16)$ , and an independent brute force for  $n = 2$ ,  $k \leq 9$ . This computation is evidence, not part of the formalized result.

## What does not follow

The proofs use the binary setting throughout: the expansion in  $x^\varepsilon(x^2 + x)^\varepsilon$  with squarefree  $x^\varepsilon$ , the additivity of  $Y \mapsto Y^{2^s}$ , the two shifts  $\delta_i \in \{0, 1\}$  in Lemma 4.2, and the telescoping identity. We make no claim about other fields or grids.

## How this paper was produced

The results were developed in a public research notebook [Notebook], whose dated entries record the exact computations that suggested the formulas, the failed routes and the reviews, using *Noemesis*, an open framework for AI-assisted mathematical research distributed with the repository [Repo]. Proposition A.2 was drafted with GPT-6 Astra in the Codex agent [Note]. The rest of the mathematics, the Lean formalization and the draft of this paper were produced by Claude Opus 5.5 in Claude Code under the author's direction, and fresh-context Claude reviewers checked the informal arguments. The author chose the problem, directed the investigation and designed the framework. The paper has not been externally peer reviewed; formal verification covers the statements with [Lean] links, not the prose or the comparison with the literature. Original manuscript material is distributed under the [CC BY 4.0](#) license.

## A A second proof of the value of $D(n, k)$

This appendix gives the proof of the lower half of Corollary 1.3 from version 1 of this paper. It does not use Section 4, and it does not determine  $\delta(n, k, \ell)$  for individual orders. Its lower bound

combines the multiplicity Schwartz–Zippel lemma with a one-dimension reduction; the upper bound is Theorem 3.7, or the hyperplane covers for  $k \geq 2^{n-1}$ . The second formal proof of Lemma 5.1 uses this route.

## A.1 The Schwartz–Zippel bound

The only result used here without proof is the multiplicity Schwartz–Zippel lemma.

**Theorem A.1** (Multiplicity Schwartz–Zippel [DKSS13, Lemma 2.7]). *Let  $\mathbb{F}$  be a field,  $\sigma$  a nonempty finite set,  $P \in \mathbb{F}[x_\sigma]$  a nonzero polynomial and  $S \subseteq \mathbb{F}$  a finite set. Then*

$$\sum_{a \in S^\sigma} \text{mult}_a(P) \leq \deg P \cdot |S|^{|\sigma|-1}$$

[Lean].

The Lean proof follows the induction on the number of variables in [DKSS13] (Lemma 8 of the arXiv version). It also proves the scaled form  $|S| \sum_a \text{mult}_a(P) \leq \deg P \cdot |S|^{|\sigma|}$ , which holds with no variables.

**Proposition A.2.** *Let  $n \geq 1$  and  $k \geq 0$ . Every nonzero  $P \in \mathbb{F}_2[x_1, \dots, x_n]$  with  $\text{mult}_a(P) \geq k$  for all  $a \neq \mathbf{0}$  satisfies*

$$\deg P \geq 2k - \lfloor k/2^{n-1} \rfloor$$

[Lean]. *In particular this holds for every  $k$ -admissible  $P$ .*

*Proof.* Put  $d = \deg P$  and  $M = 2^{n-1}$ . The  $2^n - 1$  nonzero points contribute at least  $k(2^n - 1)$  to  $\sum_{a \in \mathbb{F}_2^n} \text{mult}_a(P)$ , and Theorem A.1 with  $S = \mathbb{F}_2$  bounds this sum by  $dM$ . So  $k(2M - 1) \leq dM$ .

Let  $c = \lfloor k/M \rfloor$ , and suppose  $d + c < 2k$ . Then  $dM + (c + 1)M \leq 2kM$ . Since  $(c + 1)M > k$ , this gives  $dM < 2kM - k = k(2M - 1)$ , a contradiction.  $\square$

This is the deduction of the earlier note [Note], stated without the restriction  $k \geq 2^{n-2}$  of [BBDM23, Question 4.3]. As the note observed, the origin condition is used only to exclude  $P = 0$ .

## A.2 Linear substitutions

For a matrix  $A \in \mathbb{F}^{\sigma \times \sigma}$  over any field, let  $P \mapsto P \circ A$  be the algebra map with  $x_i \mapsto \sum_j A_{ij} x_j$ . The same matrix acts on points by  $(Aa)_i = \sum_j A_{ij} a_j$ .

**Lemma A.3.** *Let  $A, B \in \mathbb{F}^{\sigma \times \sigma}$  and  $P \in \mathbb{F}[x_\sigma]$ .*

1. *If  $P$  is homogeneous of degree  $d$ , so is  $P \circ A$ . Hence  $P \mapsto P \circ A$  commutes with taking homogeneous components, and  $\deg(P \circ A) \leq \deg P$ .*
2.  *$(P \circ A) \circ B = P \circ (AB)$ .*
3. *If  $A$  is invertible, then  $\text{mult}_{\mathbf{0}}(P \circ A) = \text{mult}_{\mathbf{0}}(P)$  and  $\text{mult}_a(P \circ A) = \text{mult}_{Aa}(P)$  for every  $a$ .*

[Lean]

*Proof.* Part 1 holds because each  $x_i$  goes to a linear form. Part 2 can be checked on the variables. For part 3, by part 2 an invertible  $A$  gives an injective substitution. The origin order is the least  $j$  with a nonzero homogeneous component of degree  $j$ , and by part 1 this is preserved.

For the second claim, the shift commutes with substitution:  $(P \circ A)(a + z) = P(Aa + Az) = (P(Aa + z)) \circ A$ . So  $\text{mult}_a(P \circ A) = \text{mult}_0(P(Aa + z) \circ A) = \text{mult}_0(P(Aa + z)) = \text{mult}_{Aa}(P)$ .  $\square$

**Lemma A.4.** Write  $\lambda_v = \sum_i v_i x_i$  for the linear form of a vector  $v$ .

1. Let  $L \in \mathbb{F}_2[x_\sigma]$  be nonzero with  $\deg L < 2^{|\sigma|} - 1$ . Then  $\lambda_v \nmid L$  for some nonzero  $v \in \mathbb{F}_2^\sigma$ .
2. Over any field  $\mathbb{F}$ , for every nonzero  $v \in \mathbb{F}^\sigma$  and every  $i_0 \in \sigma$  there is an invertible  $A$  with  $\lambda_v \circ A = x_{i_0}$ .

[Lean]

*Proof.* (1) The ring  $\mathbb{F}_2[x_\sigma]$  is a unique factorization domain. Each nonzero  $\lambda_v$  has degree one, so it is irreducible, hence prime. The only unit of  $\mathbb{F}_2[x_\sigma]$  is 1, so distinct forms are not associated. Distinct vectors give distinct forms. If all  $2^{|\sigma|} - 1$  nonzero forms divided  $L$ , their product would divide  $L$ , forcing  $\deg L \geq 2^{|\sigma|} - 1$ .

(2) By definition  $\lambda_v \circ A = \lambda_{vA}$ , with  $vA$  the row vector  $v$  times  $A$ . Extend the isomorphism between the lines  $\mathbb{F}v$  and  $\mathbb{F}e_{i_0}$  that sends  $v$  to  $e_{i_0}$  to a linear automorphism  $g$  of  $\mathbb{F}^\sigma$ , and let  $A$  be the transpose of the matrix of  $g$ . Then  $vA = g(v) = e_{i_0}$ .  $\square$

### A.3 The one-dimension step

Write the variable set as  $\sigma \cup \{x_0\}$  and a point as  $(c, a')$  with  $c \in \mathbb{F}_2$  and  $a' \in \mathbb{F}_2^\sigma$ . For  $Q \in \mathbb{F}_2[x_0, x_\sigma]$  and  $c \in \mathbb{F}_2$ , let  $Q(c, x') \in \mathbb{F}_2[x_\sigma]$  denote the substitution  $x_0 = c$ .

**Lemma A.5.** For all  $Q$ ,  $c$  and  $a'$ ,  $\text{mult}_{a'}(Q(c, x')) \geq \text{mult}_{(c, a')}(Q)$  [Lean].

*Proof.* The shift of  $Q(c, x')$  at  $a'$  is  $Q((c, a') + (w, z'))$  with  $w = 0$ . Its coefficients are the coefficients of  $Q((c, a') + (w, z'))$  at monomials without  $w$ , and these have the same degree.  $\square$

**Theorem A.6** (One-dimension step). Let  $\sigma$  be a finite set,  $n = |\sigma| + 1$  and  $k < 2^n$ . Let  $P \in \mathbb{F}_2[x_0, x_\sigma]$  satisfy  $\text{mult}_a(P) \geq k$  for every nonzero point  $a$  and  $\text{mult}_0(P) < k$ . Then there is  $S \in \mathbb{F}_2[x_\sigma]$  with  $\text{mult}_{a'}(S) \geq k$  for every nonzero  $a'$ , with  $\text{mult}_0(S) = \text{mult}_0(P)$ , and with

$$\deg S + 1 \leq \deg P$$

[Lean].

In particular  $D(n - 1, k) \leq D(n, k) - 1$  for  $n \geq 2$  and  $k < 2^n$ . Since the origin order is kept exactly, the same holds for the least degree with a prescribed origin order. By Corollary 1.3 and Theorem 1.2, both inequalities hold for every  $k \geq 1$  [Lean: D] [Lean:  $\delta$ ].

*Proof.* Let  $\ell = \text{mult}_0(P) < k$  and let  $L \neq 0$  be the homogeneous component of  $P$  of degree  $\ell$ .

*Choosing coordinates.* We have  $\deg L = \ell \leq k - 1 < 2^n - 1$ , so Lemma A.4 gives a nonzero form  $\lambda_v \nmid L$  and an invertible  $A$  with  $\lambda_v \circ A = x_0$ . Put  $Q = P \circ A$ . By Lemma A.3:

- $Q$  satisfies the hypotheses, because  $A$  permutes the nonzero points;
- $\text{mult}_0(Q) = \ell$  and  $\deg Q \leq \deg P$ ;

- the degree- $\ell$  component of  $Q$  is  $L \circ A$ ;
- $x_0 \nmid L \circ A$ : otherwise substituting  $A^{-1}$  would give  $\lambda_v \mid L$ .

*The difference.* Let  $S = Q(0, x') + Q(1, x')$ . Write  $Q = \sum_{t \geq 0} x_0^t Q_t$  with  $Q_t \in \mathbb{F}_2[x_\sigma]$ . Then  $S = \sum_{t \geq 1} Q_t$ , since the  $t = 0$  terms cancel in characteristic two.

- *Degree.*  $\deg Q_t + t \leq \deg Q$ , so  $\deg S \leq \deg Q - 1$ . Moreover  $\deg Q \geq 1$ , for otherwise  $S = 0$ , which the next item excludes.
- *Nonzero points.* For  $a' \neq \mathbf{0}$ , both  $(0, a')$  and  $(1, a')$  are nonzero, so Lemma A.5 and Lemma 2.2(2) give  $\text{mult}_{a'}(S) \geq k$ .
- *Origin order, lower bound.* Lemma A.5 gives  $\text{mult}_{\mathbf{0}}(Q(0, x')) \geq \ell$ , and  $\text{mult}_{\mathbf{0}}(Q(1, x')) \geq k$  because  $(1, \mathbf{0}) \neq \mathbf{0}$ . So  $\text{mult}_{\mathbf{0}}(S) \geq \ell$ .
- *Origin order, upper bound.* The constant term of  $L \circ A$  in  $x_0$ , namely  $(L \circ A)(0, x')$ , is nonzero because  $x_0 \nmid L \circ A$ . Take a monomial  $z^\beta$  with a nonzero coefficient in it. Then  $|\beta| = \ell$ , and the same coefficient appears in  $Q(0, x')$ . The coefficient of  $z^\beta$  in  $Q(1, x')$  vanishes, because  $\ell < k \leq \text{mult}_{\mathbf{0}}(Q(1, x'))$ . So  $z^\beta$  occurs in  $S$ , and  $\text{mult}_{\mathbf{0}}(S) \leq \ell$ .

Finally  $\deg S + 1 \leq \deg Q \leq \deg P$ . □

#### A.4 The lower bound

**Theorem A.7.** *Let  $k \geq 1$  and  $n \geq \lfloor \log_2 k \rfloor + 2$ . Every  $k$ -admissible  $P \in \mathbb{F}_2[x_1, \dots, x_n]$  satisfies*

$$\deg P \geq 2k + n - 2 - \lfloor \log_2 k \rfloor$$

[Lean].

*Proof.* Let  $n_0 = \lfloor \log_2 k \rfloor + 2$ , so  $k < 2^{n_0-1}$ . Argue by induction on  $n \geq n_0$ .

For  $n = n_0$ , Proposition A.2 gives  $\deg P \geq 2k - \lfloor k/2^{n_0-1} \rfloor = 2k$ , which is the claim.

For  $n + 1 > n_0$ , identify the variables with  $\{x_0\} \cup \sigma$ , where  $|\sigma| = n$ . This relabelling preserves degree, and by Lemma 2.2(5) it preserves multiplicities at the relabelled points. Since  $k < 2^{n_0-1} \leq 2^{n+1}$ , Theorem A.6 gives a  $k$ -admissible  $S$  in  $n$  variables with  $\deg S + 1 \leq \deg P$ . The induction hypothesis bounds  $\deg S$ . □

#### A.5 The value of $D(n, k)$

For  $k \geq 2^{n-1}$ , version 1 used the hyperplane covers of [BBDM23] as the upper bound, and the formal proof still does.

**Theorem A.8** (Hyperplane covers [BBDM23, Theorem 1.2(a) with  $d = 1$ , arXiv v1]). *Let  $n \geq 1$  and  $k \geq 2^{n-1}$ , and write  $k = a2^{n-1} + b$  with  $0 \leq b < 2^{n-1}$ . Then*

$$P = F^a \cdot (x_1^2 + x_1)^b$$

*is  $k$ -admissible and  $\deg P \leq 2k - \lfloor k/2^{n-1} \rfloor$  [Lean].*

In the language of [BBDM23],  $P$  is the product of the linear forms of a cover by  $a$  copies of the  $2^n - 1$  hyperplanes  $u \cdot x = 1$ , which avoid the origin, and  $b$  copies of the parallel pair  $x_1 = 0$ ,  $x_1 = 1$ . Their theorem gives the least size of a cover for every  $k \geq 2^{n-2}$ ; we use only this upper bound for  $k \geq 2^{n-1}$ , in the polynomial form above, which is what the Lean file proves. It follows from Lemma 3.6: the multiplicity at  $a \neq \mathbf{0}$  is at least  $a2^{n-1} + b = k$ , the origin order is below  $k$ , and the degree is at most  $a(2^n - 1) + 2b = 2k - a$ .

**Theorem A.9** (Second proof of the value of  $D(n, k)$ ). *For all  $n \geq 1$  and  $k \geq 1$ ,  $D(n, k)$  takes the values of Corollary 1.3. More precisely:*

1. for  $n \geq \lfloor \log_2 k \rfloor + 2$ , equivalently  $k < 2^{n-1}$ , every  $k$ -admissible  $P$  has  $\deg P \geq 2k + n - 2 - \lfloor \log_2 k \rfloor$ ;
2. for every  $n \geq 1$ , some  $k$ -admissible  $P$  has  $\deg P \leq 2k + n - 2 - \lfloor \log_2 k \rfloor$ ;
3. for  $k \geq 2^{n-1}$ ,  $D(n, k) = 2k - \lfloor k/2^{n-1} \rfloor$ .

[Lean: parts 1, 2] [Lean: part 3]

*Proof.* Part 1 is Theorem A.7. For  $k \geq 1$ ,  $k < 2^{n-1}$  is equivalent to  $\lfloor \log_2 k \rfloor < n - 1$ , which is  $n \geq \lfloor \log_2 k \rfloor + 2$ .

For part 2, let  $J = \lfloor \log_2 k \rfloor$  and  $\ell = k - 2^J < k$ , so that  $k - \ell - 1 = 2^J - 1$ . By Proposition 3.8 the polynomial  $y_1^\ell g_{2^J}$  is  $k$ -admissible, with degree at most  $n + 2k - 2 - s_2(2^J - 1) = n + 2k - 2 - J$  by Lemma 3.1(4).

For part 3, Proposition A.2 gives the lower bound and Theorem A.8 the upper bound.  $\square$

## B Formal-verification map and reproduction

The arguments are complete in the body; reading them requires no Lean. This appendix locates the formal declarations and gives commands to check them. File paths are relative to `formalization/`. The table uses the prefix `MR.` for `MathResearch.`; the external inputs lie under `MathResearch.ThirdParty`.

| Paper statement                                  | Lean file and declarations                                                                                                                                                            |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hasse multiplicity,<br>Definition 2.1; Lemma 2.2 | <a href="#">claims/HasseMultiplicity.lean</a><br>MR.HasseMultiplicity.mult, coe_le_mult_iff, min_le_mult_add,<br>add_le_mult_mul, mult_mul, mult_eq_zero_iff,<br>mult_rename_equiv    |
| Digit sums, Lemma 3.1                            | <a href="#">claims/TruncatedProduct.lean</a><br>MR.TruncatedProduct.s2_add_le, s2_two_pow, s2_le_self,<br>two_mul_add_s2_le;<br>MR.BinaryMultiplicityDegreeFormula.s2_two_pow_sub_one |
| Legendre form, Lemma 3.2                         | <a href="#">claims/PerOrderLegendre.lean</a><br>MR.PerOrderLegendre.legendre, phi_of_lt_two_pow, bits,<br>length_bits, sum_bits, lt_of_mem_bits                                       |
| Digit-sum identities used in the<br>remarks      | <a href="#">claims/BinaryDigitSums.lean</a><br>MR.BinaryDigitSums.s2_mul_two_pow_add, two_pow_s2_le,<br>s2_le_log, s2_pos, s2_eq_one_iff, s2_succ_add_padicValNat                     |

| Paper statement                                                                                                                                                                                                                                             | Lean file and declarations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Catalan parity [AK73, DS06],<br>Section 3                                                                                                                                                                                                                   | <a href="#">third-party-claims/CatalanParity.lean</a><br>MR.ThirdParty.CatalanParity.padicValNat_catalan,<br>odd_catalan_iff                                                                                                                                                                                                                                                                                                                                                                                     |
| Telescoping, Lemma 3.3;<br>truncated product, Theorem 3.5                                                                                                                                                                                                   | <a href="#">claims/TruncatedProduct.lean</a><br>MR.TruncatedProduct.telescope, g, truncated_product                                                                                                                                                                                                                                                                                                                                                                                                              |
| Hyperplane product, Lemma 3.6;<br>construction, Theorem 3.7                                                                                                                                                                                                 | <a href="#">claims/PerOrderUpperBound.lean</a><br>MR.PerOrderUpperBound.hypProd, mult_hypProd,<br>eval_zero_hypProd, totalDegree_hypProd, Phi, construction,<br>construction_spec, per_order_upper_bound                                                                                                                                                                                                                                                                                                         |
| Menezes' polynomial,<br>Proposition 3.8                                                                                                                                                                                                                     | <a href="#">claims/PerOrderConstruction.lean</a><br>MR.PerOrderConstruction.per_order_construction                                                                                                                                                                                                                                                                                                                                                                                                               |
| Expansion, Lemma 4.1                                                                                                                                                                                                                                        | <a href="#">claims/PerOrderExpansion.lean</a><br>MR.PerOrderExpansion.exists_expansion,<br>weight_le_totalDegree                                                                                                                                                                                                                                                                                                                                                                                                 |
| Artin-Schreier root, Section 4                                                                                                                                                                                                                              | <a href="#">claims/ArtinSchreierSeries.lean</a><br>MR.ArtinSchreierSeries.asRoot, coeff_asRoot_eq_one_iff,<br>asRoot_sq_add, trunc_asRoot                                                                                                                                                                                                                                                                                                                                                                        |
| Congruence, Lemma 4.2                                                                                                                                                                                                                                       | <a href="#">claims/PerOrderYCoefficient.lean</a><br>MR.PerOrderYCoefficient.xhat, xhat_sq_add, expansion_congr,<br>coeff_expansion                                                                                                                                                                                                                                                                                                                                                                               |
| Forms, Lemma 4.4                                                                                                                                                                                                                                            | <a href="#">claims/PerOrderForms.lean</a><br>MR.PerOrderForms.E, E_perm, E_span, E_eval, E_rec                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Grid vanishing, Lemma 4.5; grid<br>reduction, Lemma 4.6                                                                                                                                                                                                     | Mathlib.MvPolynomial.eq_zero_of_eval_zero_at_prod_finset;<br><a href="#">claims/PerOrderSubspaceReduction.lean</a><br>MR.PerOrderSubspaceReduction.rho, coeff_reduction                                                                                                                                                                                                                                                                                                                                          |
| Lowest part, Lemma 4.3; lower<br>bound, Theorem 4.7                                                                                                                                                                                                         | <a href="#">claims/PerOrderLowerBound.lean</a><br>MR.PerOrderLowerBound.wser, mult_wser, hc_wser_ne, cond,<br>E_vanish, lower_bound                                                                                                                                                                                                                                                                                                                                                                              |
| Main theorem, Theorem 1.2                                                                                                                                                                                                                                   | <a href="#">claims/PerOrderValue.lean</a><br>MR.PerOrderValue.per_order_value                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Minimum over the origin order,<br>Lemma 5.1; extremal orders,<br>Remark 5.3                                                                                                                                                                                 | <a href="#">claims/PerOrderMinimumDirect.lean</a><br>MR.PerOrderMinimumDirect.hval, hval_le, hval_attain,<br>min_phi_direct, phi_eq_degreeMin_iff,<br>extremal_of_le_two_pow, extremal_example, delta_eq_D_iff,<br>delta_eq_D_of_le_two_pow, delta_extremal_example;<br><a href="#">claims/PerOrderMinimum.lean</a> MR.PerOrderMinimum.min_phi                                                                                                                                                                   |
| Arithmetic of $\Phi$ : Section 1<br>(Legendre form, comparison<br>with Menezes' form, blocks,<br>examples, the top order, the<br>reals), Section 2 (substitutions),<br>Section 3 (exact degree, the<br>factors of $F$ ), Appendix A (one<br>more dimension) | <a href="#">claims/PerOrderArithmetic.lean</a><br>MR.PerOrderArithmetic.phi_add_eq, phi_lt_menezes,<br>construction_lt_menezes, phi_add_block, phi_of_le,<br>phi_of_real_range, phi_top, phi_succ,<br>totalDegree_construction, menezes_form_optimal, one_le_s2,<br>example_one, cube_example, example_two,<br>mult_le_mult_zero_subst, card_nonzero, mult_hypProd_sq,<br>totalDegree_hypProd_sq; for $\delta$ itself: deltaSet, isLeast_deltaSet,<br>delta_succ, delta_of_real_range, delta_top, delta_examples |

| Paper statement                                                | Lean file and declarations                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arithmetic of $D(n, k)$ :<br>Sections 1 and 5, Appendix A      | <a href="#">claims/DegreeFormulaArithmetic.lean</a><br>MR.DegreeFormulaArithmetic.degreeMin_one, degreeMin_four,<br>degreeMin_eight, degreeMin_fifteen, degreeMin_excess,<br>degreeMin_succ, degreeMin_three_mul, degreeMin_add_one_le;<br>for $D$ itself: degreeSet, isLeast_degreeSet, admissible_ne_zero,<br>D_examples, D_excess, D_succ, D_three_mul, D_add_one_le |
| Value of $D(n, k)$ , Corollary 1.3;<br>Theorem A.9(3)          | <a href="#">claims/BinaryMultiplicityDegreeComplete.lean</a><br>MR.BinaryMultiplicityDegreeComplete.degreeMin,<br>degree_complete, cover_bound_attained_iff                                                                                                                                                                                                             |
| Extended range, Corollary 5.2                                  | <a href="#">claims/BinaryMultiplicityDegreeExtendedRange.lean</a><br>MR.BinaryMultiplicityDegreeExtendedRange.degree_formula_<br>extended                                                                                                                                                                                                                               |
| Multiplicity Schwartz–Zippel<br>[DKSS13], Theorem A.1          | <a href="#">third-party-claims/MultiplicitySchwartzZippel.lean</a><br>MR.ThirdParty.MultiplicitySchwartzZippel.multiplicity_<br>schwartz_zippel                                                                                                                                                                                                                         |
| Schwartz–Zippel bound over $\mathbb{F}_2$ ,<br>Proposition A.2 | <a href="#">claims/BinaryMultiplicityCoverBound.lean</a><br>MR.BinaryMultiplicityCoverBound.cover_bound,<br>cover_bound_of_origin                                                                                                                                                                                                                                       |
| Linear substitutions,<br>Lemma A.3; restriction,<br>Lemma A.5  | <a href="#">claims/OneDimensionStep.lean</a><br>MR.OneDimensionStep.homogeneousComponent_subst,<br>totalDegree_subst_le, subst_comp, mult_zero_subst,<br>mult_subst, mult_restrict                                                                                                                                                                                      |
| Avoiding a linear form,<br>Lemma A.4                           | <a href="#">claims/LinearFormAvoidance.lean</a><br>MR.LinearFormAvoidance.exists_linForm_not_dvd,<br>exists_subst_linForm_eq_X                                                                                                                                                                                                                                          |
| One-dimension step,<br>Theorem A.6                             | <a href="#">claims/OneDimensionStep.lean</a><br>MR.OneDimensionStep.one_dimension_step                                                                                                                                                                                                                                                                                  |
| Lower bound, Theorem A.7                                       | <a href="#">claims/BinaryMultiplicityLowerBound.lean</a><br>MR.BinaryMultiplicityLowerBound.lower_bound                                                                                                                                                                                                                                                                 |
| Hyperplane covers [BBDM23],<br>Theorem A.8                     | <a href="#">third-party-claims/HyperplaneCoverUpperBound.lean</a><br>MR.ThirdParty.HyperplaneCoverUpperBound.hyperplane_cover_<br>upper_bound, card_dot_eq_one                                                                                                                                                                                                          |
| Second proof, Theorem A.9(1,2);<br>Definition 1.1              | <a href="#">claims/BinaryMultiplicityDegreeFormula.lean</a><br>MR.BinaryMultiplicityDegreeFormula.Admissible,<br>degree_formula, degree_formula_exact,<br>degree_formula_of_lt_two_pow                                                                                                                                                                                  |

## Reading the formal statements.

- Points are functions  $\text{Fin } n \rightarrow \mathbb{Z}\text{Mod } 2$ , degree is `MvPolynomial.totalDegree`, and multiplicity is `MR.HasseMultiplicity.mult`, the order of  $P(a+z)$  in  $\mathbb{N}_\infty$ .  $\Phi$  is `MR.PerOrderUpperBound.Phi`, written out with natural-number division.
- Section 4 uses one polynomial ring for both  $x$  and  $y$ :  $A_\varepsilon(x^2 + x)$  is `aeval yv (A ε)`. The series  $\hat{x}_K$  is `xhat K`, and  $w$  is `wser k P`, with  $K = k$ .
- In `MR.PerOrderForms.E`, the coefficient  $w_{k-1-t-\Sigma}$  is written `hcw k w (t + Σ)`, which is zero when  $t + \Sigma \geq k$ .

- The one-dimension step is stated for the variables  $\text{Option } \sigma$ , with none playing the role of  $x_0$ . The lower bound of Appendix A relabels  $\text{Fin}(n + 1)$  as  $\text{Option}(\text{Fin } n)$ .

**Commands.** The project pins Lean 4.34.0-rc2 and Mathlib in `lakefile.lean` and `lake-manifest.json`. With [elan](#), Git and Python 3 installed:

```
git clone https://github.com/kbr-/math-research.git bmd-check
cd bmd-check
git checkout --detach 168a0c8eef547709b4c0a18115ff97b01c20200d
cd formalization
export MATHLIB_NO_CACHE_ON_UPDATE=1
lake exe cache get claims/BinaryMultiplicityDegreePaper2.lean
python3 verify.py --target claims/BinaryMultiplicityDegreePaper2.lean \
--out verification.txt
```

The target is an import-only module whose header lists 64 declarations: the main ones (`per_order_value`, `construction_spec`, both `lower_bound` theorems, `min_phi`, `min_phi_direct`, `degree_complete`, `cover_bound_attained_iff`, `per_order_construction`) and those behind the remarks and comparisons. It imports every module in the table, so the build covers the whole chain, and its axiom report is transitive. The verifier builds the proofs, prints each listed declaration’s type, and rejects unfinished proofs and nonstandard axioms. The expected axiom report for the main theorem is

```
'MathResearch.PerOrderValue.per_order_value'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

and the same for every other listed declaration. These are Lean’s standard foundations.

**Kernel replay.** To recheck the compiled proofs, and every imported declaration including Mathlib’s, in a fresh Lean kernel, run

```
lake env leanchecker --fresh claims.BinaryMultiplicityDegreePaper2
```

from `formalization/`. On 25 September 2026 this passed on the author’s machine in 204 seconds, on sources identical to the pinned commit.

**Independent validation.** All builds and replays reported here ran on the author’s machine. No hosted or third-party build of this module has been recorded; the repository’s Lean workflow currently targets a different preprint.

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